An Inversion-Preserving Bijection Between Cayley Permutations and Fubini Rankings

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Combinatorics and Coding Theory in the Tropics REU 2026

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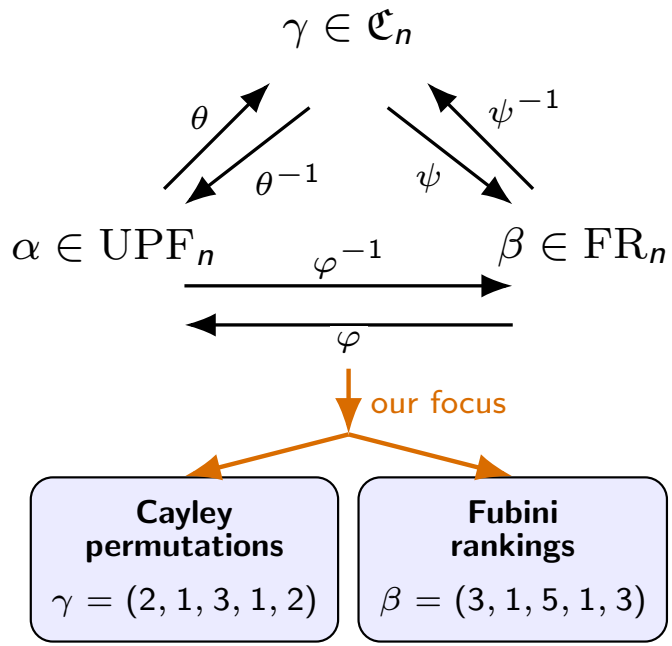
Outline

- Motivation and definitions
- Question
- Result

Motivation

- We studied three families. Our focus here will be on Cayley permutations and Fubini rankings.

Three families in the paper



Cayley Permutations

A Cayley permutation of length n is a tuple

$$\gamma = (c_1, \ldots, c_n) \in [n]^n$$

whose distinct values are $1, 2, \ldots, m$ for some $m \leq n$.

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There can be no missing values before the largest one.
Repeated values are allowed.

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$$\gamma = (2, 1, 3, 1, 2)$$

The values that appear are $\{1, 2, 3\}$.

No values are missing, so
γ is a Cayley permutation.

Cayley Permutation Check

Are the following Cayley permutations?

1. $(1, 1, 1, 1, 1)$

Cayley Permutation Check

Are the following Cayley permutations?

1. $(1, 1, 1, 1, 1)$ Yes

Cayley Permutation Check

Are the following Cayley permutations?

1. $(1, 1, 1, 1, 1)$ Yes
2. $(4, 2, 1, 3, 2)$

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Cayley Permutation Check

Are the following Cayley permutations?

1. $(1, 1, 1, 1, 1)$ Yes
2. $(4, 2, 1, 3, 2)$ Yes
3. $(3, 1, 3, 1, 3)$

Cayley Permutation Check

Are the following Cayley permutations?

1. $(1, 1, 1, 1, 1)$ Yes
2. $(4, 2, 1, 3, 2)$ Yes
3. $(3, 1, 3, 1, 3)$ No

It is not a Cayley permutation because the value 2 is missing.

Fubini Rankings

A Fubini ranking of length n is a tuple

$$\beta = (\beta_1, \ldots, \beta_n) \in [n]^n$$

in which rank 1 appears, and if a rank i occurs k times, then the next larger rank that appears is $i + k$.

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If k competitors tie at rank i, then the next rank is $i + k$.
Ties(repeats) are allowed.

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$$\beta = (3, 1, 5, 1, 3)$$

Rank 1 occurs twice, so the next rank is $1 + 2 = 3$.
Rank 3 occurs twice, so the next rank is $3 + 2 = 5$.

Therefore, β is a Fubini ranking.

Fubini Ranking Check

Are the following Fubini rankings?

1. $(1, 2, 3, 4, 5)$

Fubini Ranking Check

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Fubini Ranking Check

Are the following Fubini rankings?

1. $(1, 2, 3, 4, 5)$ Yes
2. $(1, 1, 3, 3, 5)$ Yes
3. $(1, 1, 1, 3, 5)$ No

It is not a Fubini ranking because after three 1's, the next rank must be 4.





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An **Inversion** -Preserving **Bijection** Between Cayley Permutations and Fubini Rankings





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Bijections

A function $f : A \to B$ is a bijection if it is both *one-to-one* and *onto*.

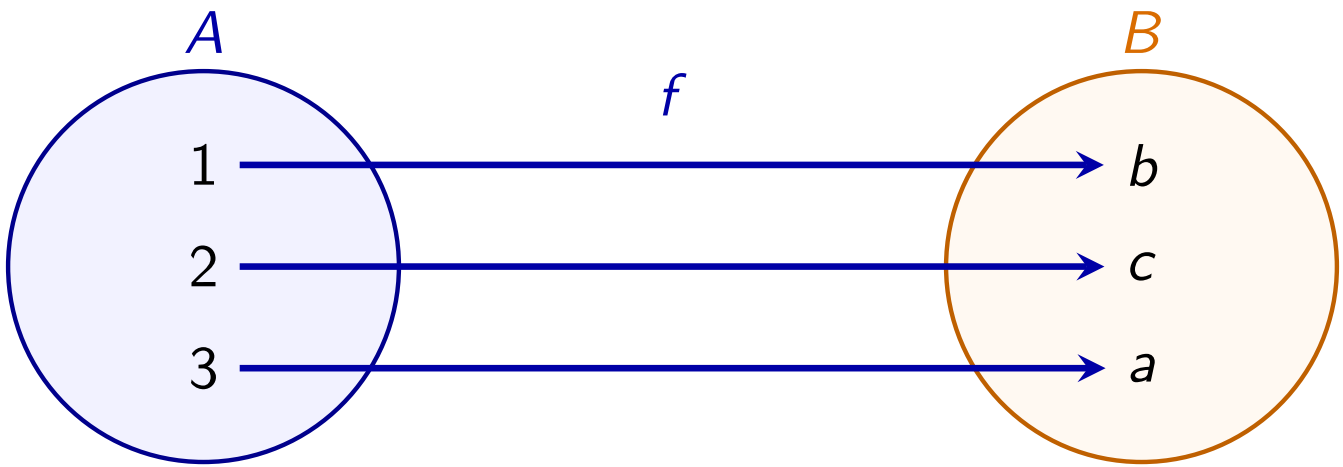
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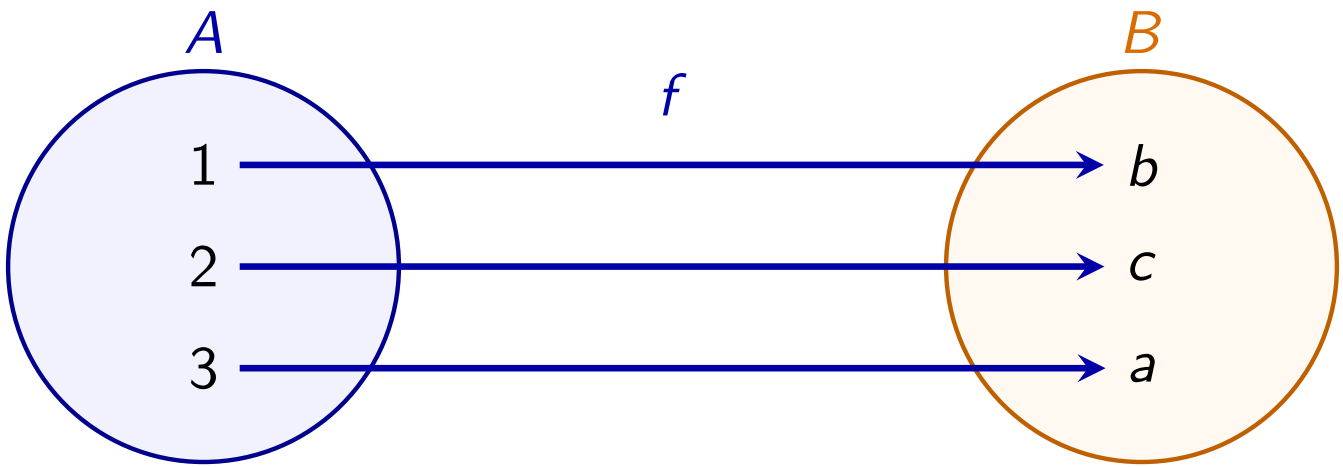


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In words: Every object in B is paired with *exactly one* object in A.

Example



No two inputs share an output, and no output is missed.

Therefore, f is a bijection and $|A| = |B|$.

Inversions

Let $w = (w_1, \ldots, w_n)$. An inversion is a pair of positions (i, j) such that $i < j$ but $w_i > w_j$.

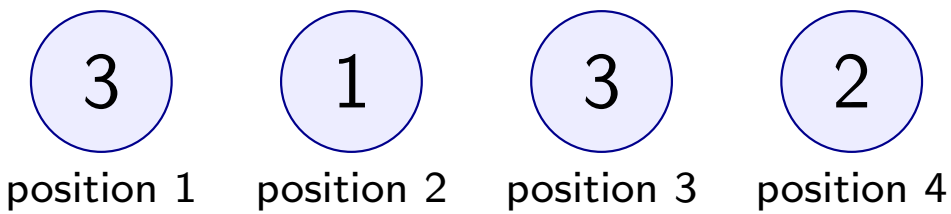
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Example: $w = (3, 1, 3, 2)$

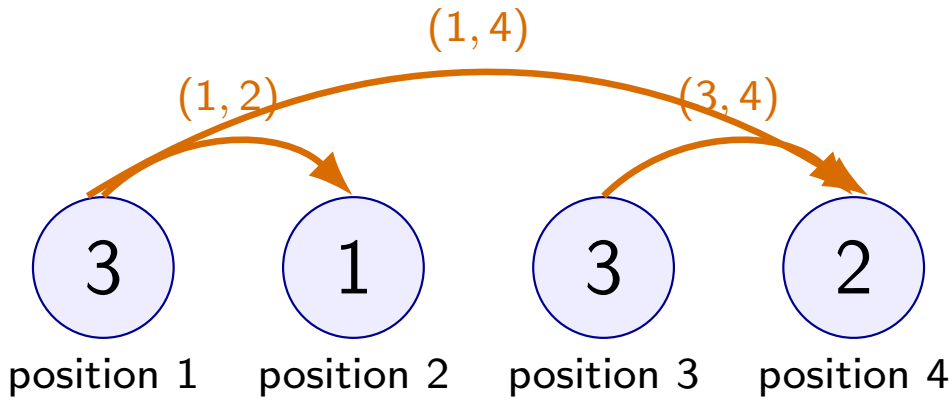


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Each arrow goes from a larger entry to a smaller entry on its right.

$$\mathrm{Inv}(w) = \{(1,2), (1,4), (3,4)\}, \qquad \mathrm{inv}(w) = 3.$$

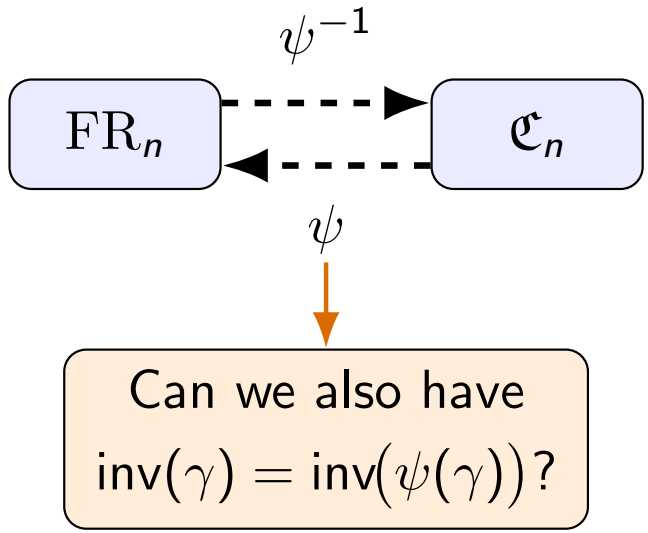
The two equal entries 3 and 3 do not form an inversion.

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- We studied three families. Our focus here will be on Cayley permutations and Fubini rankings.
- Can we construct a bijection

$$\psi : \mathfrak{C}_n \longrightarrow \mathrm{FR}_n$$

that preserves inversions?



Question

Can we construct a bijection

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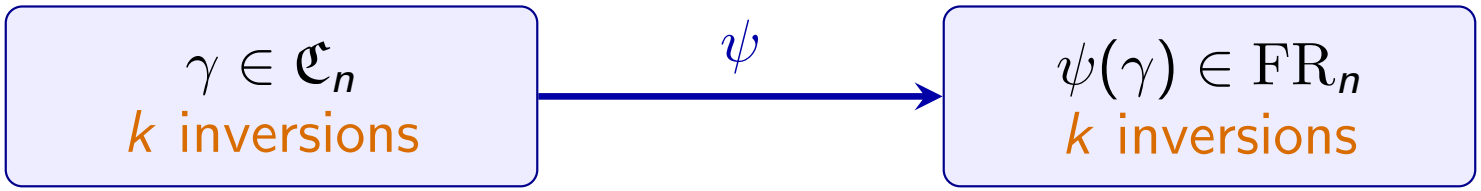
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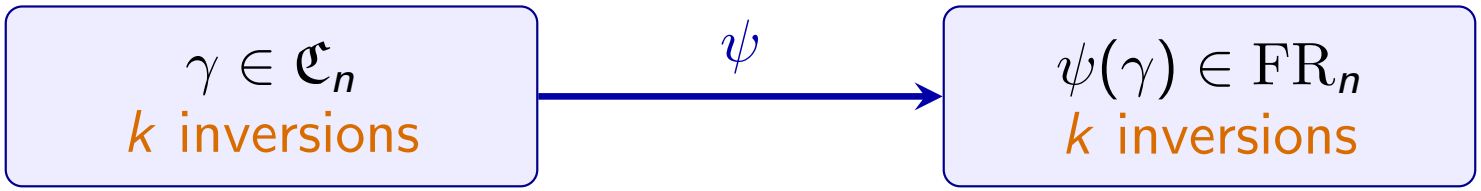
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Suppose $\gamma \in \mathfrak{C}_n$ has k inversions.



In words: Applying ψ may change the tuple, but it does not change how many pairs are out of order.

$$\mathrm{inv}(\gamma) \quad = \quad \mathrm{inv}(\psi(\gamma)).$$

Main Result





Theorem

For any $n \in \mathbb{N}$, the sets $\mathfrak{C}_n$ and FR_n are in bijection.

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Lemma 1

Given $\gamma = (c_1, c_2, \ldots, c_n) \in \mathfrak{C}_n$, define $g : \mathfrak{C}_n \to \mathrm{FR}_n$ by $g(\gamma) = (b_1, b_2, \ldots, b_n)$, where $b_i = 1 + |\{1 \leq j \leq n : c_j < c_i\}|$. for every $i \in [n]$. The map g is well defined.

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Lemma 2

Given $\beta = (b_1, b_2, \ldots, b_n) \in \mathrm{FR}_n$, let $r_1, r_2, \ldots, r_m$ be the $m \leq n$ distinct ranks appearing in β, where $r_1 < r_2 < \cdots < r_m$. Define $\psi : \mathrm{FR}_n \to \mathfrak{C}_n$ by $\psi(\beta) = (c_1, c_2, \ldots, c_n)$, where $c_i = j$ whenever $b_i = r_j$. The map ψ is well defined.

$\mathfrak{C}_n$ to FR_n mapping

Start with a Cayley permutation

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For each entry c_i:

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$$\boxed{(\gamma)_i = \psi(\gamma)_i = 1 + \left|\{j \in [n] : c_j < c_i\}\right|}$$

Example

$$\gamma = (2, 1, 3, 1, 2) \in \mathfrak{C}_5$$

position i	1	2	3	4	5
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$$(2, 1, 3, 1, 2) \xrightarrow{\psi} (3, 1, 5, 1, 3)$$

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FR_n to $\mathfrak{C}_n$ Mapping

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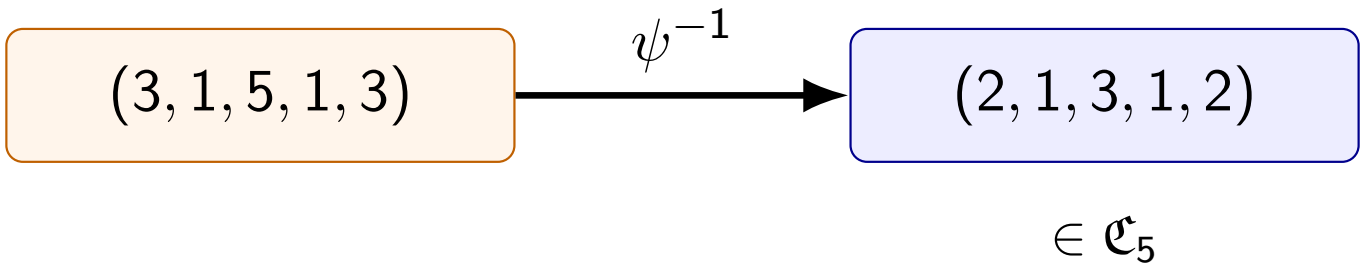
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$$(3, 1, 5, 1, 3) \xrightarrow{\psi^{-1}} (2, 1, 3, 1, 2)$$

$$\in \mathfrak{C}_5$$

This undoes the other mapping, returning the Cayley permutation we started with.

Main Result

Theorem

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Lemma 2 ✓

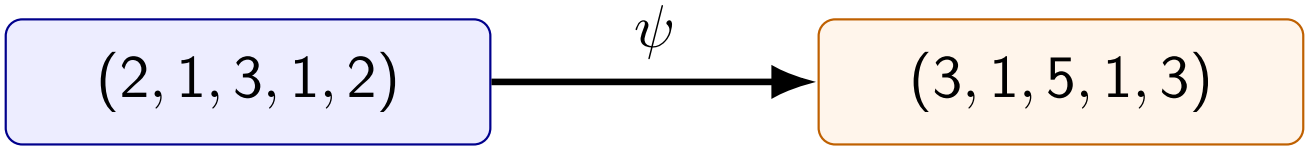
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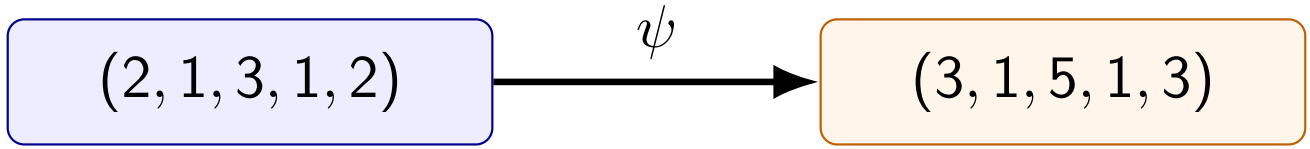


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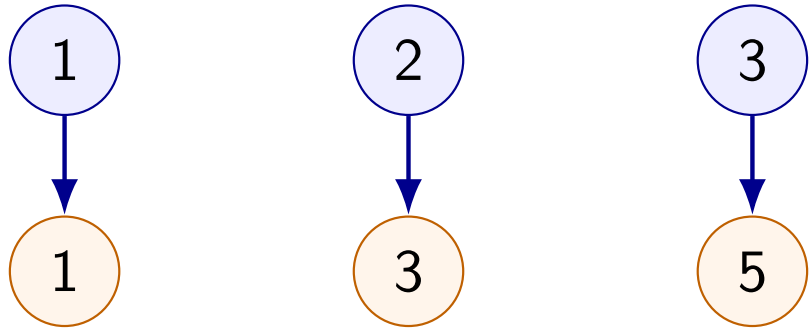
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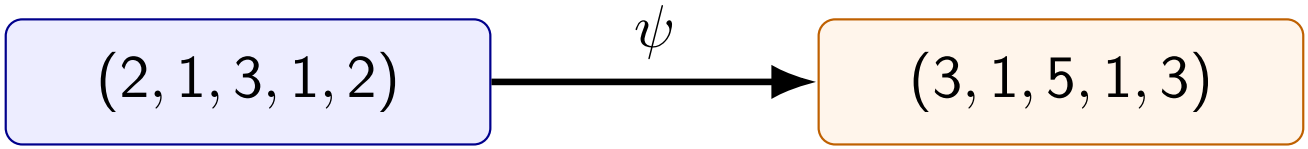
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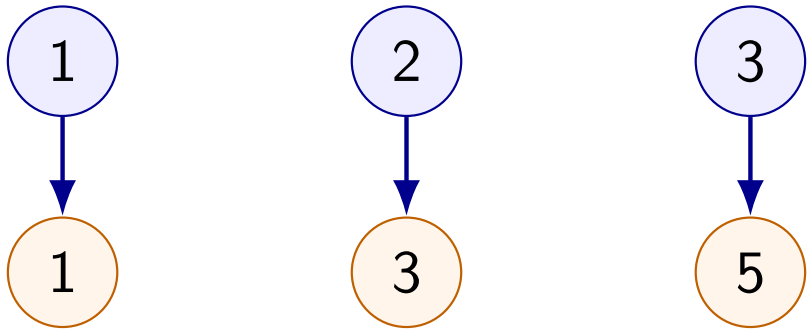
The values change like this:



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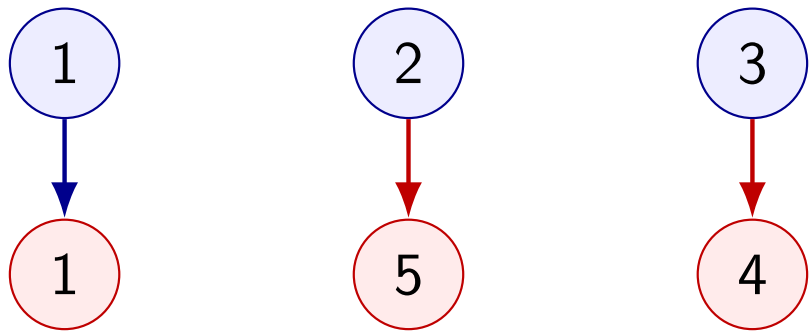
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Order is preserved: $1 < 2 < 3$ becomes $1 < 3 < 5$.

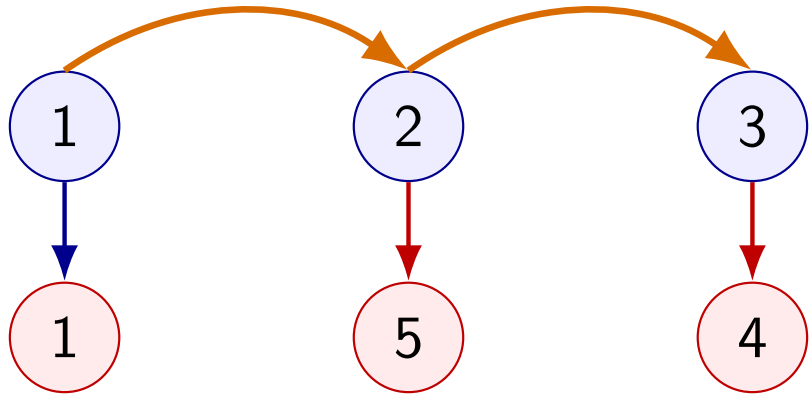
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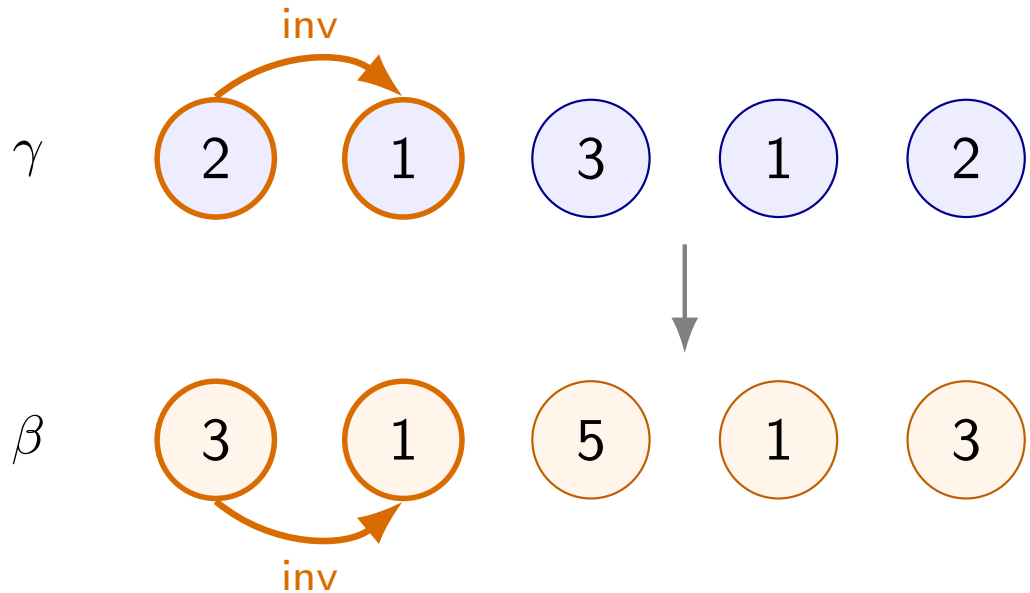


$1 < 5$, but not $5 < 4$

Order fails, so inversions would not be preserved.

How Are Inversions Preserved?

Returning to our map, look at positions $(1, 2)$:



$2 > 1$ becomes $3 > 1$, still an inversion.

How Are Inversions Preserved?

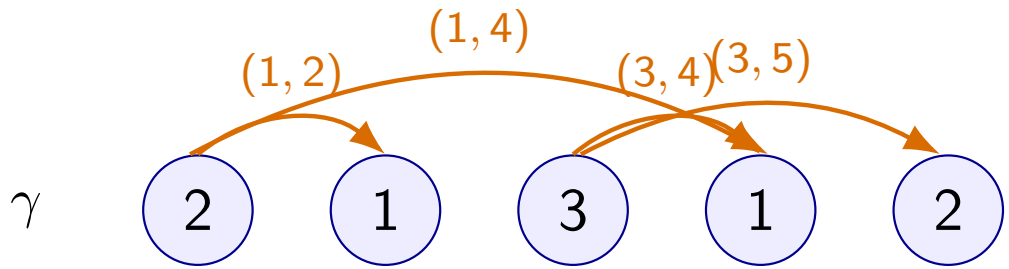
Both tuples have the same inversion pairs:

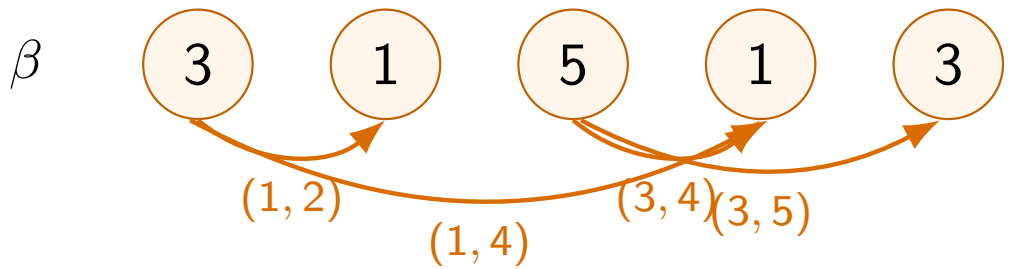
γ (2) (1) (3) (1) (2)

β (3) (1) (5) (1) (3)

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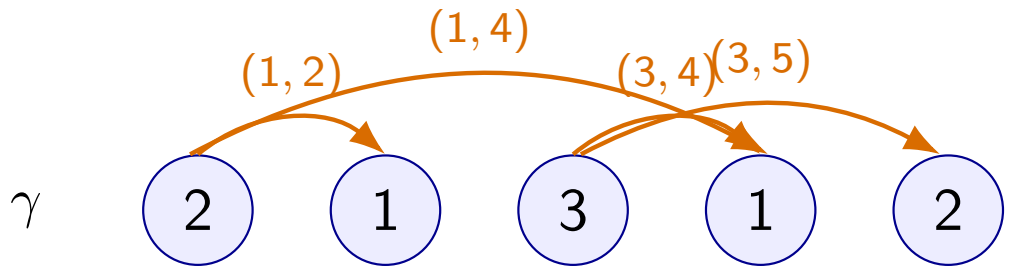
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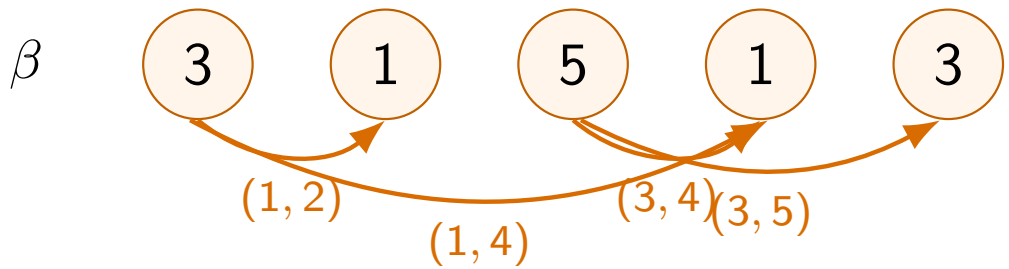




How Are Inversions Preserved?

Both tuples have the same inversion pairs:





$$\text{Inv} = \{(1,2),(1,4),(3,4),(3,5)\}, \qquad \text{inv} = 4.$$

Summary





We constructed an explicit inversion-preserving bijection between Cayley permutations and Fubini rankings.

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$$(2,1,3,1,2) \in \mathfrak{C}_5 \xrightarrow{\ \psi\ } (3,1,5,1,3) \in \mathrm{FR}_5$$

inv $= 4$ on both sides

Thank You!



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Research Colleague



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References

[1] Bjorn Andreas Ager-Hart, Melissa Beerbower, Pamela E. Harris, Joakim Jakovleski, and Matt McClinton.
Counting ℓ-interval fubini rankings through their parking outcome, 2026.
arXiv:2603.00693.

[2] S. Alex Bradt, Jennifer Elder, Pamela E. Harris, Gordon Rojas Kirby, Eva Reutercrona, Yuxuan Wang, and Juliet Whidden.
Unit interval parking functions and the r-Fubini numbers.
La Matematica, 3(1):370–384, 2024.

[3] Kyle Celano, Jennifer Elder, Kimberly P. Hadaway, Pamela E. Harris, Jeremy L. Martin, Amanda Priestley, and Gabe Udell.
Statistics on ℓ-interval parking functions, 2025.
arXiv:2507.07243.

[4] Giulio Cerbai.
Sorting cayley permutations with pattern-avoiding machines, 2020.
arXiv:2003.02536.

[5] Anders Claesson, Giulio Cerbai, Dana C. Ernst, and Hannah Golab.
Pattern-avoiding cayley permutations via combinatorial species, 2024.
arXiv:2407.19583.

[6] Jennifer Elder, Pamela E. Harris, Jan Kretschmann, and J. Carlos Martínez Mori.
Parking functions, fubini rankings, and boolean intervals in the weak order of $\mathfrak{S}_n$.
Journal of Combinatorics, 16(1):65–89, 2025.